

Compliance progress

1 Progress Update: Issuer Schema Reconciliation and Table-Aware Extraction

This round of development focused on stabilizing the issuer schema inference and reconciliation pipeline by explicitly incorporating table-aware evidence extraction, enforcing strict type discipline, and reusing high-precision reconcile-phase evidence to seed issuer instances. The changes materially improve correctness, auditability, and downstream compatibility with Lean-based verification.

1.1 Motivation

Earlier iterations of the pipeline suffered from three systemic issues: (i) ambiguity over whether extracted values originated from narrative text or structured tables, (ii) incorrect type inference caused by naive numeric parsing (e.g., Indian comma formats), and (iii) schema instability due to unconstrained LLM judge outputs when evidence was sparse or missing. These issues propagated downstream, negatively impacting issuer extraction and Lean code generation.

1.2 Key Improvements Implemented

Table-Aware Evidence Extraction A table-first extraction strategy was enforced for rule-anchored fields. Structured financial tables (e.g., multi-year FY summaries) are now parsed explicitly and preferred over narrative text. Each extracted evidence record includes debug markers indicating whether a table was loaded and whether a table row was used, removing ambiguity during evaluation and regression testing.

Robust Numeric Parsing Numeric parsing was centralized into a deterministic utility capable of handling Indian comma formats and decimals (e.g., `1,44,445.3`). Commas are stripped, decimals are handled deterministically, and lists are derived exclusively from table columns rather than naive comma splitting. This eliminated spurious list inference and stabilized numeric types.

Strict Type Enforcement The reconciliation phase now enforces a closed set of allowed schema types (e.g., `Nat`, `List Nat`, `Option String`). Invalid or hallucinated types produced by the judge (such as `"missing"` or malformed option types) are deterministically repaired or downgraded. Fields with no supporting evidence are explicitly marked inactive and typed conservatively.

Controlled Judge Behavior The LLM judge is no longer permitted to infer non-optional or structured types when the missing rate is 100%. In such cases, deterministic overrides force safe fallback types, preventing speculative schema inflation and improving Lean compatibility.

Issuer Candidate Emission A new artifact, `issuer_candidate.json`, is generated directly from the reconciliation phase. This file aggregates high-confidence, rule-anchored evidence into a partial issuer instance with full provenance. Table-derived values are prioritized, conflicts are preserved explicitly, and all values conform to the reconciled issuer schema.

1.3 Current Capabilities

At the end of this round, the pipeline reliably:

- Extracts multi-year financial fields from tables as `List Nat`.
- Produces a stable, Lean-compatible issuer schema.
- Seeds issuer instances using high-precision reconcile-phase evidence.
- Preserves provenance and conflict information for auditability.

1.4 Intended Downstream Use

The reconciled schema and issuer candidate can now be safely consumed by: (i) schema-guided issuer extraction passes for completeness, (ii) Lean code generation for issuer structures and rules, and (iii) formal compliance verification using rule-specific check functions.

This establishes a clear separation between high-precision rule-anchored extraction and completeness-oriented issuer population, improving both correctness and maintainability.

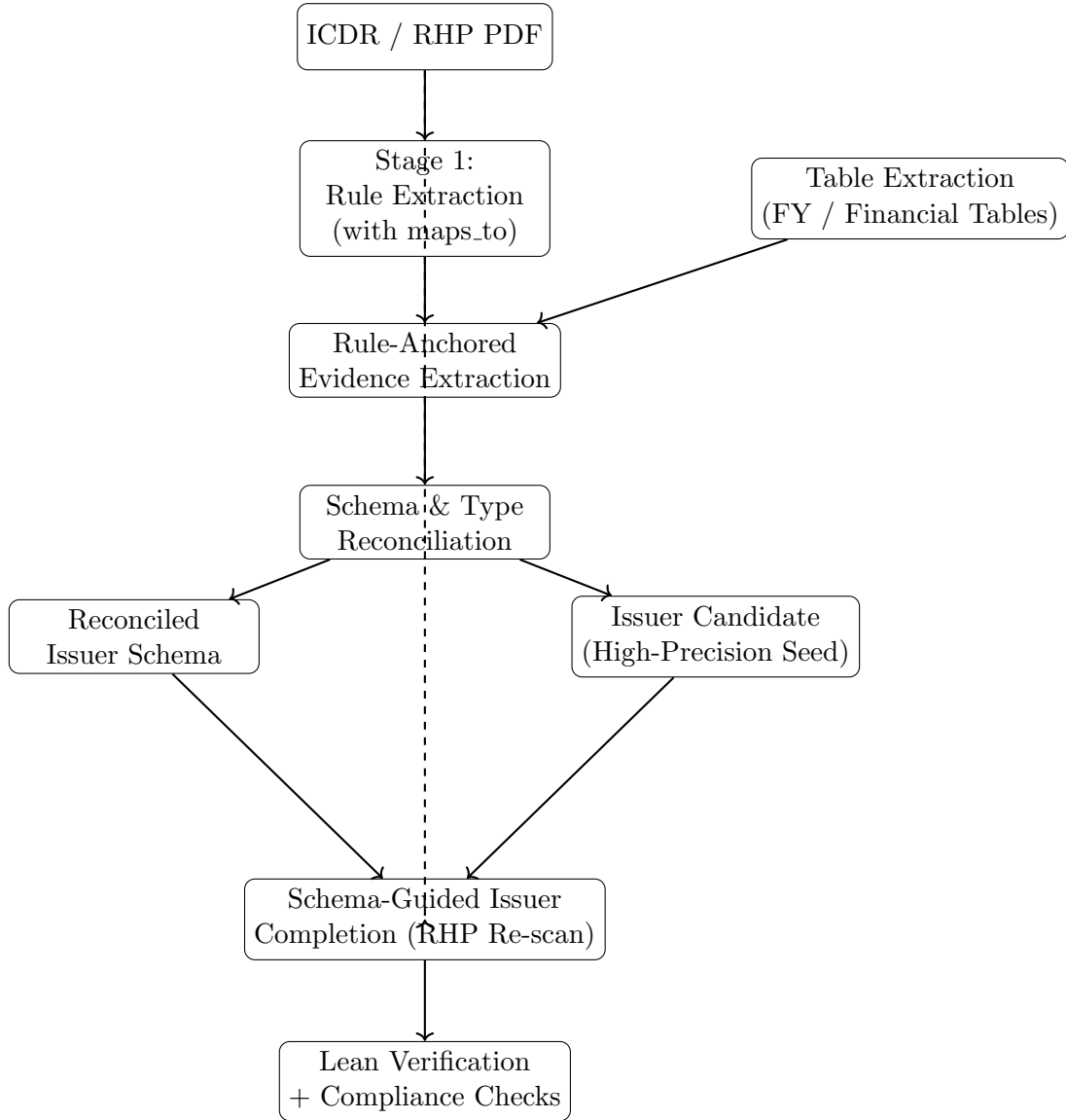


Figure 1: Updated pipeline showing reuse of the RHP after reconciliation for schema-guided issuer completion.